

MONG CAI, Viet Nam

WMO: 488380

Lat: 21.517N

Lon: 107.967E

Elev: 7

StdP: 101.24

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	9.6	-0.2	3.7	12.5	2.2	4.4	12.8	6.4	13.8	5.9	14.5	1.4	50	0.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	33.0	27.5	32.4	27.4	31.9	27.3	29.0	31.4	28.6	31.1	28.2	30.7	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.3	24.5	30.7	27.9	24.0	30.4	27.5	23.5	30.1	94.9	31.6	93.0	31.2	91.3	30.8	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.1	5.1	4.3	6.5	34.5	1.2	0.6	5.6	34.9	4.8	35.3	4.1	35.6	3.3	36.1	
			4.1	29.5	1.3	0.5	3.1	29.9	2.4	30.2	1.6	30.5	0.6	30.9	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	15.4	16.6	19.6	23.7	26.8	28.6	28.7	28.5	27.8	25.4	21.5	17.2
	DBStd	5.43	3.14	3.66	3.23	2.65	1.84	1.38	1.31	1.31	1.38	2.05	2.86	2.99
	HDD10.0	3	2	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	263	99	71	25	2	0	0	0	0	0	0	7	58
	CDD10.0	4873	169	184	298	411	522	557	578	573	534	477	347	222
	CDD18.3	2092	8	22	65	163	264	307	320	315	284	218	104	22
	CDH23.3	21014	3	28	165	1015	2696	3756	3949	3781	3176	1878	531	35
CDH26.7	7209	0	0	5	153	809	1491	1596	1461	1132	504	58	1	
Wind	WSAvg	1.6	1.5	1.5	1.4	1.5	1.6	1.7	1.8	1.4	1.6	1.7	1.6	1.7
Precipitation	PrecAvg	2631	42	41	62	102	247	462	652	457	276	95	75	34
	PrecMax	3865	258	254	161	287	602	828	1488	1244	481	491	268	112
	PrecMin	1833	1	0	8	7	49	167	123	93	34	7	1	2
	PrecStd	496	53	45	40	80	135	171	312	240	118	102	74	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.9	27.3	30.3	32.4	33.6	33.7	34.0	33.2	32.2	29.2	25.6
		MCWB	19.5	22.6	24.0	25.7	27.3	27.9	28.0	27.3	26.4	25.1	23.1	21.2
	2%	DB	22.1	24.2	26.2	29.3	31.3	32.7	32.7	32.9	32.4	30.9	28.1	24.3
		MCWB	18.6	21.6	23.3	25.4	26.9	27.7	28.0	27.6	26.4	24.4	23.1	20.7
	5%	DB	21.1	22.9	25.1	28.4	30.6	32.0	32.1	32.2	31.7	30.1	27.2	23.2
		MCWB	18.5	20.7	22.8	25.1	26.7	27.8	27.9	27.5	26.3	24.2	22.7	19.6
	10%	DB	20.1	21.7	24.1	27.5	30.0	31.3	31.5	31.5	31.0	29.3	26.3	22.0
		MCWB	17.9	19.9	22.1	24.7	26.3	27.6	27.7	27.3	26.1	23.9	22.1	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	23.1	24.7	27.4	29.0	29.2	29.7	29.2	28.4	27.0	25.6	23.6
		MCDB	22.0	25.5	26.6	28.8	31.2	31.9	32.0	31.9	31.2	30.1	27.5	24.8
	2%	WB	20.2	22.1	23.8	26.4	28.1	28.7	29.0	28.6	27.8	26.2	24.9	22.2
		MCDB	21.3	23.8	25.4	28.1	30.2	31.3	31.4	31.3	30.7	29.1	26.8	23.4
	5%	WB	19.3	21.2	23.2	25.7	27.6	28.4	28.6	28.2	27.4	25.7	24.2	20.9
		MCDB	20.5	22.4	24.8	27.6	29.6	30.8	31.0	30.9	30.3	28.5	26.2	22.3
	10%	WB	18.5	20.3	22.4	25.1	27.1	28.1	28.1	27.8	27.0	25.2	23.4	19.6
		MCDB	19.8	21.4	23.8	27.0	29.2	30.4	30.6	30.5	29.9	27.8	25.4	21.3
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.5	4.0	4.3	4.7	4.5	4.5	5.0	5.7	6.5	6.5	6.3
		MCDBR	6.6	5.9	5.2	5.3	5.3	5.4	5.4	6.2	6.8	7.6	7.7	7.6
		MCWBR	3.9	3.7	3.2	2.8	2.4	2.2	2.2	2.5	2.5	3.1	3.8	4.2
	5% WB	MCDBR	4.8	4.8	4.6	4.5	4.6	4.8	4.8	5.6	5.8	6.1	5.3	5.1
		MCWBR	3.6	3.5	3.2	2.8	2.4	2.4	2.5	2.7	2.8	3.0	3.4	4.0
Clear-Sky Solar Irradiance	taub	0.608	0.644	0.807	0.759	0.561	0.533	0.508	0.570	0.615	0.712	0.598	0.622	
	taud	1.737	1.698	1.448	1.570	2.049	2.145	2.208	2.007	1.860	1.622	1.812	1.729	
	Ebn at Noon	660	670	586	628	762	776	796	751	709	611	664	628	
	Edh at Noon	213	234	312	280	172	155	146	178	204	249	196	208	
All-Sky Solar Radiation	RadAvg	2.26	2.56	2.76	3.64	4.45	4.47	4.58	4.54	4.48	4.09	3.37	2.64	
	RadStd	0.56	0.46	0.41	0.42	0.42	0.57	0.54	0.27	0.29	0.29	0.48	0.46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.46	+0.19	+0.17	N/A	N/A	N/A	+62

Nomenclature: See separate page